

Blue Spruce Park Map Revisions

Work completed by the UAV II class at IUP AGES

May 5, 2026

Dr. Schaney

Blue Spruce Park Map Revisions

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Blue Spruce Park – Taylor, Ben

Brief Background

Blue Spruce Park is a 650-acre park located 6 miles north of Indiana, PA that was established by the county in 1966. The park was originally bought by the Buffalo, Rochester, and Pittsburgh railroad during its campaign to buy watershed land to provide pure water for its locomotives. For many years, water than contained acid mine pollution was one of the railroads' largest expenses. The water contained diluted sulfuric acid, which was eaten away at the boilers and reduced the working life of the locomotives and increased maintenance costs. In response to this, the railroad built Cummins dam, named after an early landowner J.D. Cummins in 1908. It was later enlarged in 1912. The park contains historic marker dedicated to Bernice Gera, the first female umpire in baseball history. The park features over five miles of hiking trails, and there are gently sloped hills on either side of the lake that are open to beginner skiing. The park is open to hunting year-round and features a 12-acre stocked lake for fishing. The park also features 4 pavilions and two lodges that are available to the public to rent. Additionally, the park features two playgrounds, a volleyball court, horseshoe court, and open areas for sports and games. Finally, the park features the Indiana County Master Gardeners Demonstration Gardens. It was initially created in 1996 when an Indiana Girl Scout troop received a youth conservation grant to construct a butterfly garden. A large donation was additionally made by Dorthy Musser. This evolved into the gardens of today. Today's gardens feature native grasses, shrubs, and plants, a deer resistant garden, a rose garden, pollinator garden, Daylily garden, a rain garden, and a memorial garden. There is also an extensive herb garden which is grouped into uses, such as tea and culinary sections.



-Photos of the park from the early 1900's

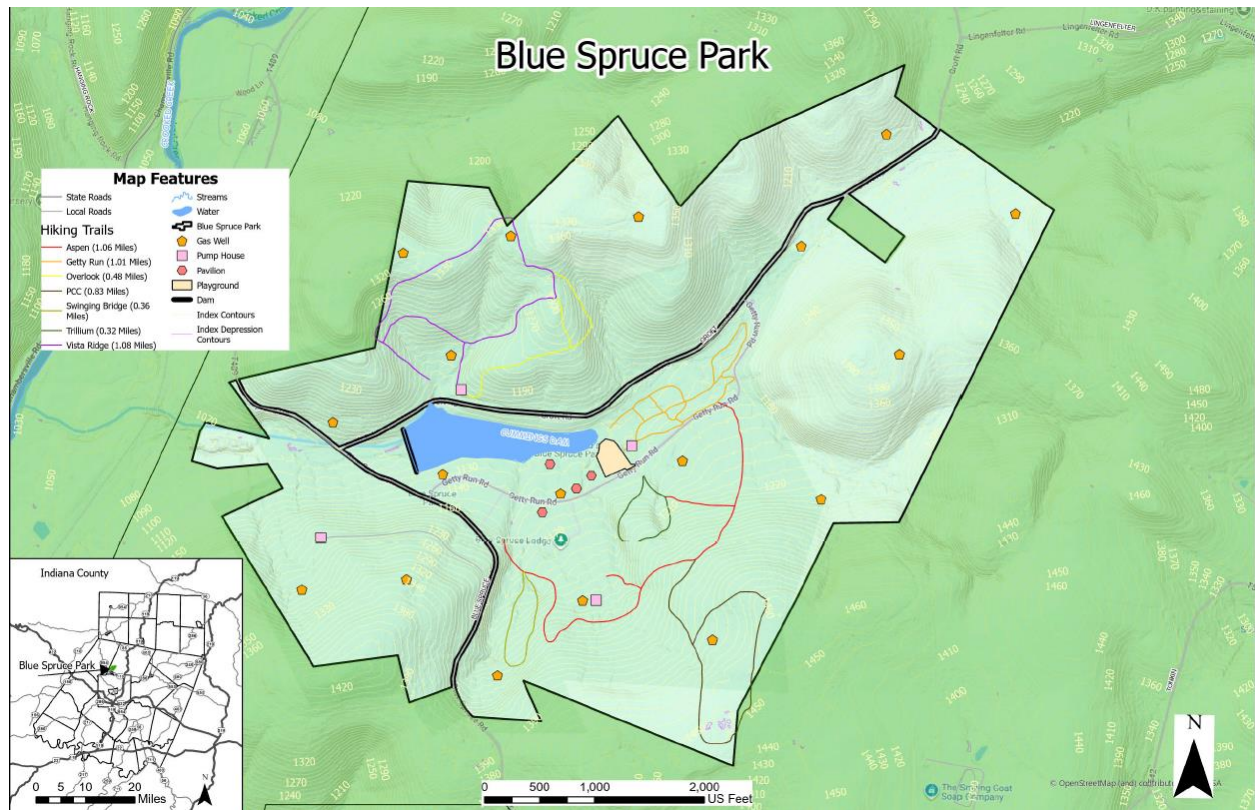
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Why are we doing this?

The purpose of the GEOG 463 course is to give students in-depth exposure to the requirements needed to enter the professional drone pilots' work industry. Students will learn to plan projects and produce deliverables such as 2D orthomosaics, digital elevation models (DEMs) and 3D models, as well as demonstrate skills required by the UAV industry. Student will gain experience of the knowledge and skills necessary to undertake manual and automated UAS flights for spatial data acquisition, such as fundamental photogrammetry concepts, UAS mission planning, GPS/GNSS ground control, UAS airborne navigation, and processing of UAV collected data into data deliverables. Students may be required to manage and administer UAS flight missions, including flight plans, crew management, risk minimization, logistics, data acquisition, and data processing for specific applications, demonstrating the ability to integrate UAS-acquired data into various geospatial outputs.

The purpose of this project is to develop more efficient and accurate maps of Blue Spruce Park in Indiana, Pennsylvania. By improving the quality and precision of geospatial data, this work supports better navigation, planning, and overall understanding of the park's layout. (See map below)

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Current map of Blue Spruce Park before starting our project

In addition, this project enhances the Indiana University of Pennsylvania (IUP) student experience by providing hands-on, field-based learning opportunities. Students gain practical experience in data collection, analysis, and mapping, allowing them to apply classroom knowledge in a real-world setting.

This project also fosters a deeper understanding of the complete reporting process, from initial data collection to final presentation. Through collaboration, students develop teamwork and communication skills while working together to gather data, organize findings, and produce a comprehensive final report.

Who are We Working for and Why?

This project is being conducted in collaboration with the Indiana County Planning Department. Key contacts for this project include Josh Krug, Deputy Director of the Planning Section, and Ed Patterson, Director, who granted permission for drone operations and data collection.

The Indiana County Planning Department is interested in obtaining updated and more accurate maps of Blue Spruce Park to support park management, planning, and future

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improvements. Enhanced geospatial data allows for better decision-making and contributes to maintaining and improving park facilities for public use.

In addition, this partnership provides mutual benefits. The department receives valuable mapping data at no cost, while students at Indiana University of Pennsylvania gain practical, real-world experience in geospatial data collection, analysis, and reporting. This collaboration supports both community development and student learning outcomes.

UAV- Maya, Caroline

What Drone are we using?

Mavic 3 Enterprise

- The DJI Mavic 3 Enterprise (M3E) series is a compact, high-performance commercial drone featuring a 4/3 CMOS 20MP sensor with a mechanical shutter for rapid mapping, 56× zoom, and optional RTK module for centimeter-level accuracy. Designed for enterprise, public safety, and inspection, the series includes the 3T (thermal) and 3M (multispectral) versions, offering 45-minute flight times. (DJI Enterprise, 2026)

Camera Specs

- The M3E includes a high-performance camera designed for mapping and inspection, centered around a 20MP 4/3 CMOS wide sensor with a mechanical shutter to eliminate rolling shutter distortion. It includes a 12MP zoom camera (up to 56× hybrid) for detailed inspection and supports an RTK module for centimeter-level precision. The M3T (Thermal) version replaces the main sensor with a 640×512 resolution thermal camera. (DJI Enterprise, 2026)

Why are we using this Drone?

- The M3E provides high mapping efficiency and accuracy, with camera features that eliminate motion blur, which is ideal for developing accurate aerial imagery of a large-scale vegetative area like Blue Spruce Park. This level of precision is optimal for ecological mapping and habitat assessment, where small spatial errors have the potential to influence the final image. The drone is also foldable, lightweight, and can be deployed instantly in the field, which allowed for convenient transportation and flight at the park. These logistical advantages reduced setup time and allowed for more consistent flight planning, overall data reliability, and uniform study site coverage.

Vector Based Data- Michael, Marina, Matthew, James

Marina:

- Better inset map
- Roads, trails, clipping

Matthew:

- Indiana county centerlines, main roads, Micropolitan Stat Area Data
- Pulled and collected useful data from S Drive
- Wrote some info points on the road, trail, county boundary and 3.2 DEM data

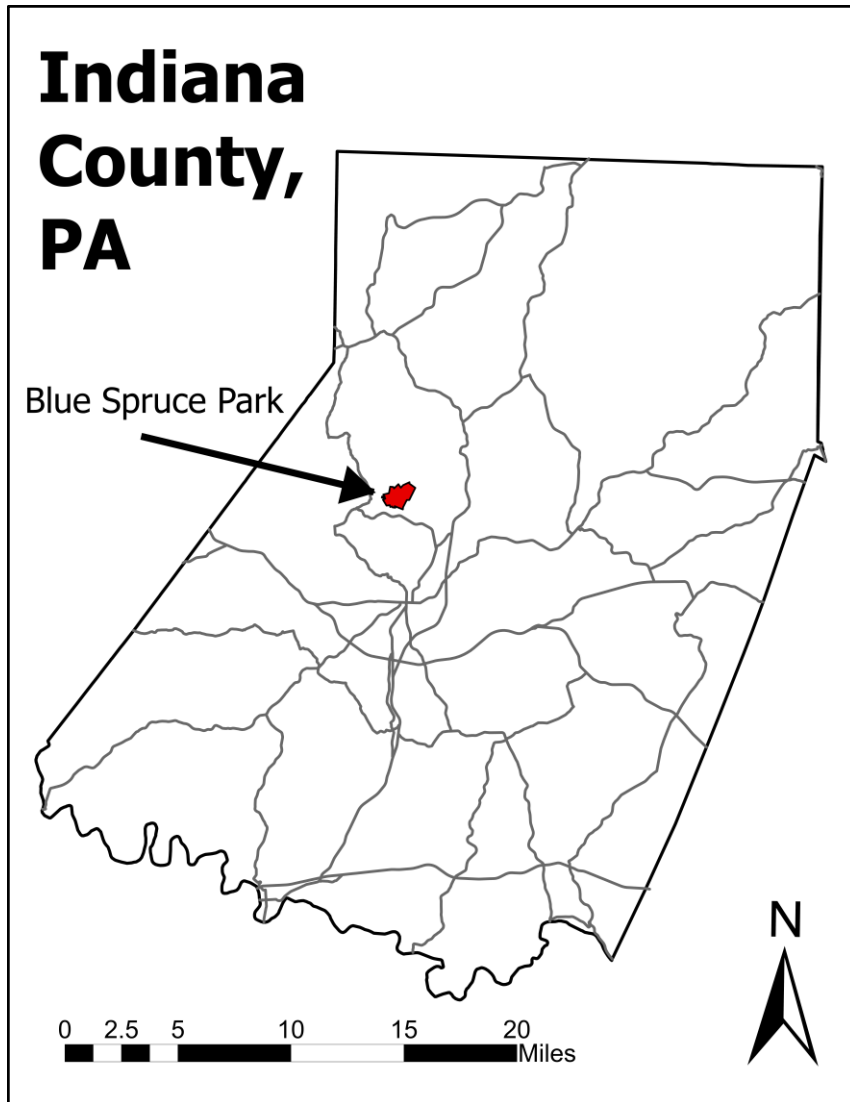
Michael:

- Primary roads, Blue Spruce Polygon, county boundary, park boundary, dock/handicap accessible, gas wells, bird interaction area, 3.2 ft DEM based in lidar, cut park to boundary
- Main map

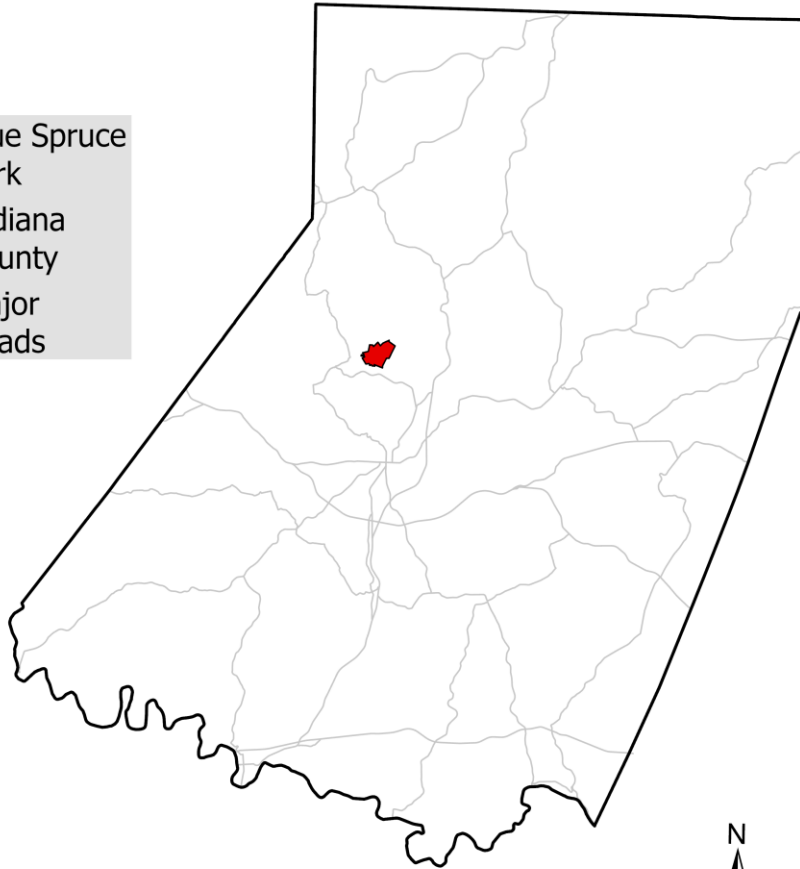
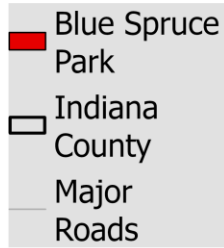
James:

- Indiana point, Cummings Reservoir, Pavilions, Restrooms, Playground, Parking

-Inset Map:



Indiana County, PA

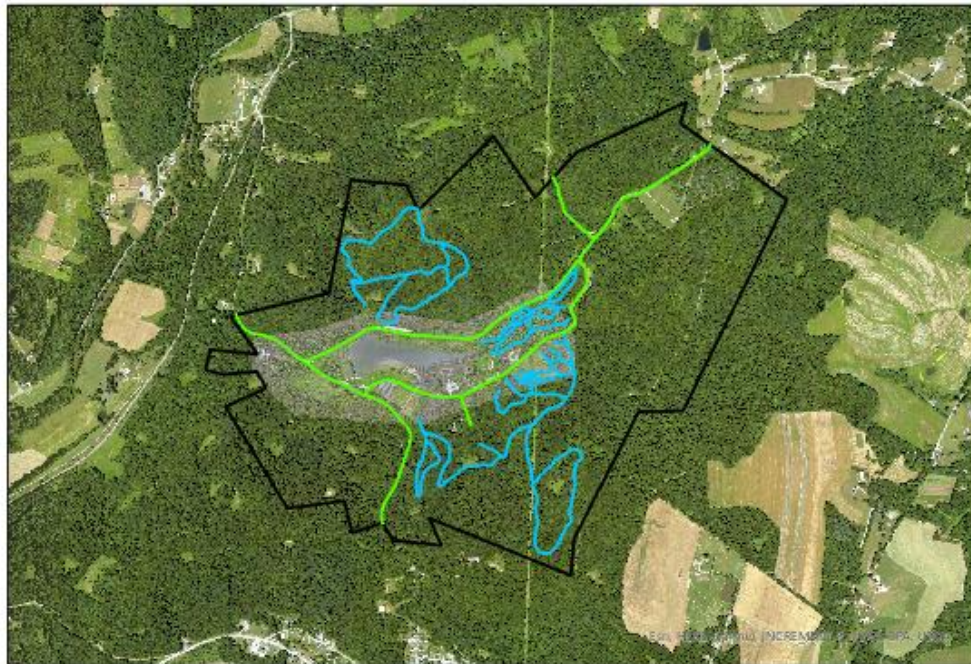


0 5 10 20 Miles



Blue Spruce Park Map Revisions

-Blue Spruce Map 1:



Legend

Blue Spruce Roads
Blue Spruce Trails

Mavic 3E Flight

RGB

Red: Band_1
Green: Band_2
Blue: Band_3

Park Boundary

Blue Spruce NAIP

RGB

Red: Band_1
Green: Band_2
Blue: Band_3

N



0 0.38 0.75 1.5 Miles

Blue Spruce Park Map Revisions

-Blue Spruce Map 2:



Legend

- | | |
|-----------------------|------------------|
| • Lakeside Center | Green: Band_2 |
| • Park Office | Blue: Band_3 |
| • Wells | ▭ Park Boundary |
| • Bird Feeding | Blue Spruce NAIP |
| • Handicap Accessible | RGB |
| • Dock | Red: Band_1 |
| • Docks | Green: Band_2 |
| Mavic 3E Flight | Blue: Band_3 |
| RGB | |
| Red: Band_1 | |



Blue Spruce Park Map Revisions

-Blue Spruce Map 3:



Legend

- Lakeside Center
- Park Office
- Wells
- Bird Feeding
- Handicap Accessible Dock
- Docks

Mavic 3E Flight

RGB

Red: Band_1

Green: Band_2

Blue: Band_3

Park Boundary

Blue Spruce NAIP

RGB

Red: Band_1

Green: Band_2

Blue: Band_3



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-Blue Spruce Map 4:



Legend

Hillshade DEM

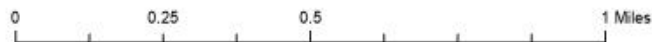
Value



Mavic 3E Flight

RGB

- Red: Band_1
- Green: Band_2
- Blue: Band_3
- Park Boundary



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- Primary Roads, line: obtained from PASDA, originating from PA Department of Conservation and Natural Resources, and Bureau of Topographic and Geologic Survey, created to be used as a general-purpose mapping of main Pennsylvania roads.
- Blue Spruce Boundary, polygon: Blue Spruce Park obtained from PASDA.
- County Boundary, polygon: Indiana County Boundary obtained from PASDA, originating from Homeland Infrastructure Foundation-Level Data – HIFLD, created to show the Pennsylvania Micropolitan Statistical Areas originally, this gave a good boundary for Indiana County.
- Dock, points: Fishing docks for water access, points created by IUP Students.
- Handicap Accessible Dock, points: Fishing dock for water access, handicap accessible, points created by IUP Students.
- Gas wells, points: Gas well locations within Blue Spruce Park, points created by IUP Students.
- Bird Interaction Area, point: Bird Interaction area on Gettysburg Trail, points created by IUP students.
- Park Office, point: Blue Spruce Park Office, point created by IUP students.
- Lakeside Center, point: Lakeside Information Center of Blue Spruce Park, point created by IUP students.
- Trails, line: Various trails of Blue Spruce Park, obtained from PASDA- Data created by Explore PA Trails. Trails - Explore PA Trails, originating from the Pennsylvania Dept. of Conservation and Natural Resources, and Bureau of State Parks, created to give official data for trails across Pennsylvania by using quality checked GPS units. The Explore PA Trails matched data collected by IUP students through GPS units.
- 3.2 Foot DEM, raster image: Digital elevation model displaying elevation of Blue Spruce Park, obtained from PASDA (PAMAP), originating from PAMAP Program, PA Department of Conservation and Natural Resources, and Bureau of Topographic and Geologic Survey, created to give a topographic aerial view of Pennsylvania, particularly for scientific, resource management, hydrologic modeling, resource modeling, general mapping, and to global asses global change.

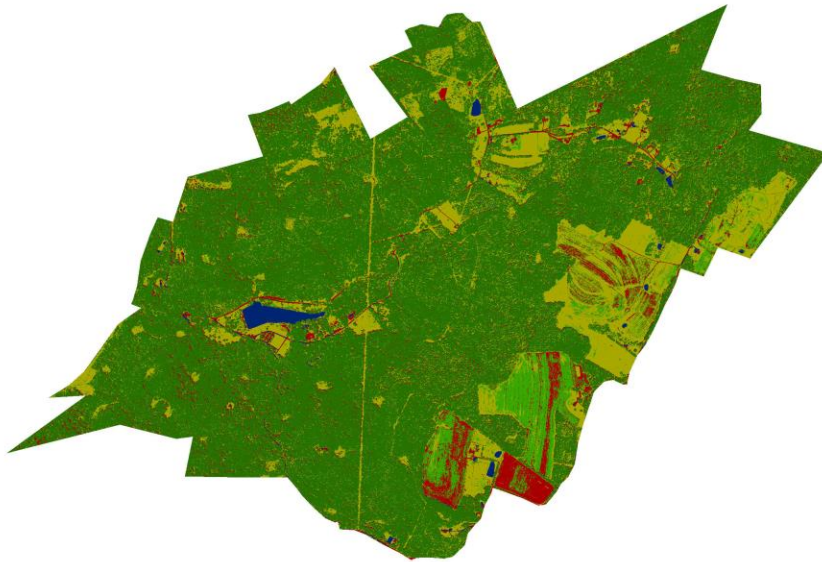
Raster Based Data- Frank, Tyler, Josh

Supervised / Unsupervised classification of recreation and conservation zone

Description of the technique

With unsupervised classification, the pixels are clustered together based on spectral homogeneity and spectral distance. Spectral homogeneity is evaluated by the software program Spectral distance is measured using various techniques chosen by the analyst. Unsupervised classification produces classes of homogeneous spectral identities that may not exactly correspond to informational classes. While spectral classes denote pixels grouped by uniform brightness values, informational classes refer to groups that convey useful information with designations such as land use, soil classes, or agricultural productivity. When using unsupervised classification, the informational class may contain a variety of spectral identities. Therefore, a spectral class identified by unsupervised classification seldom matches exactly to an informational class.

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Legend

Classes

Value

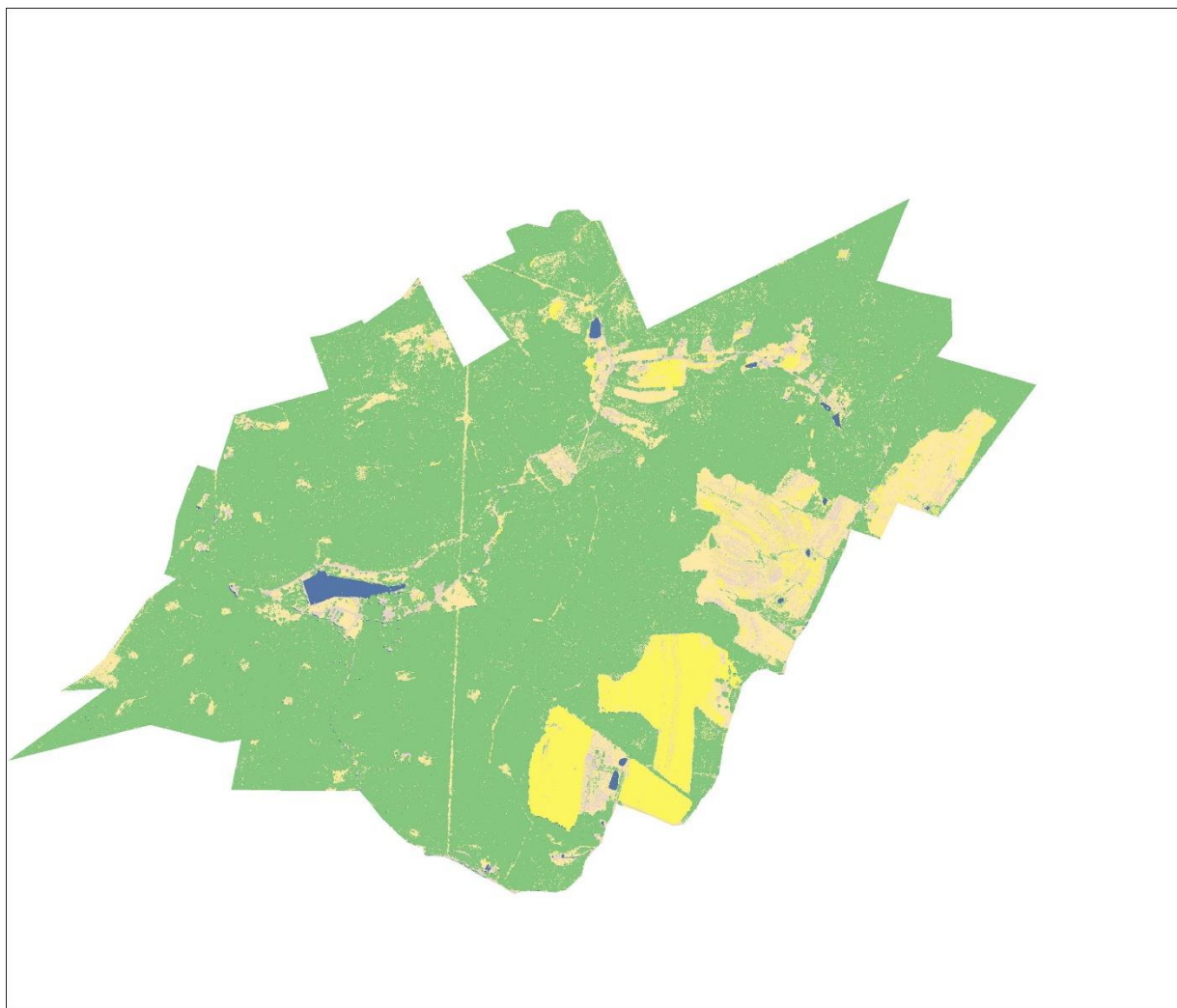
	Water
	Built
	Forest
	Used Ground
	Shrub
	BS_Rec_Conserv_Zone

An analyst has more control over the supervised classification process than the software-controlled unsupervised classification process. Through supervised classification, the analyst selects pixels representing recognized landscape features/patterns or pixels that can be identified from other sources, such as high-resolution aerial photos. Knowledge of the data, the classes desired, and the algorithm to be used is required before selecting training samples—samples of spectral values within the image which pertain to specific features/patterns. The analyst “trains” the software to identify pixels with similar

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characteristics by identifying patterns in the imagery and chooses the spectral classes for the informational classes, thus, supervising the classification process.

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Legend

Classified

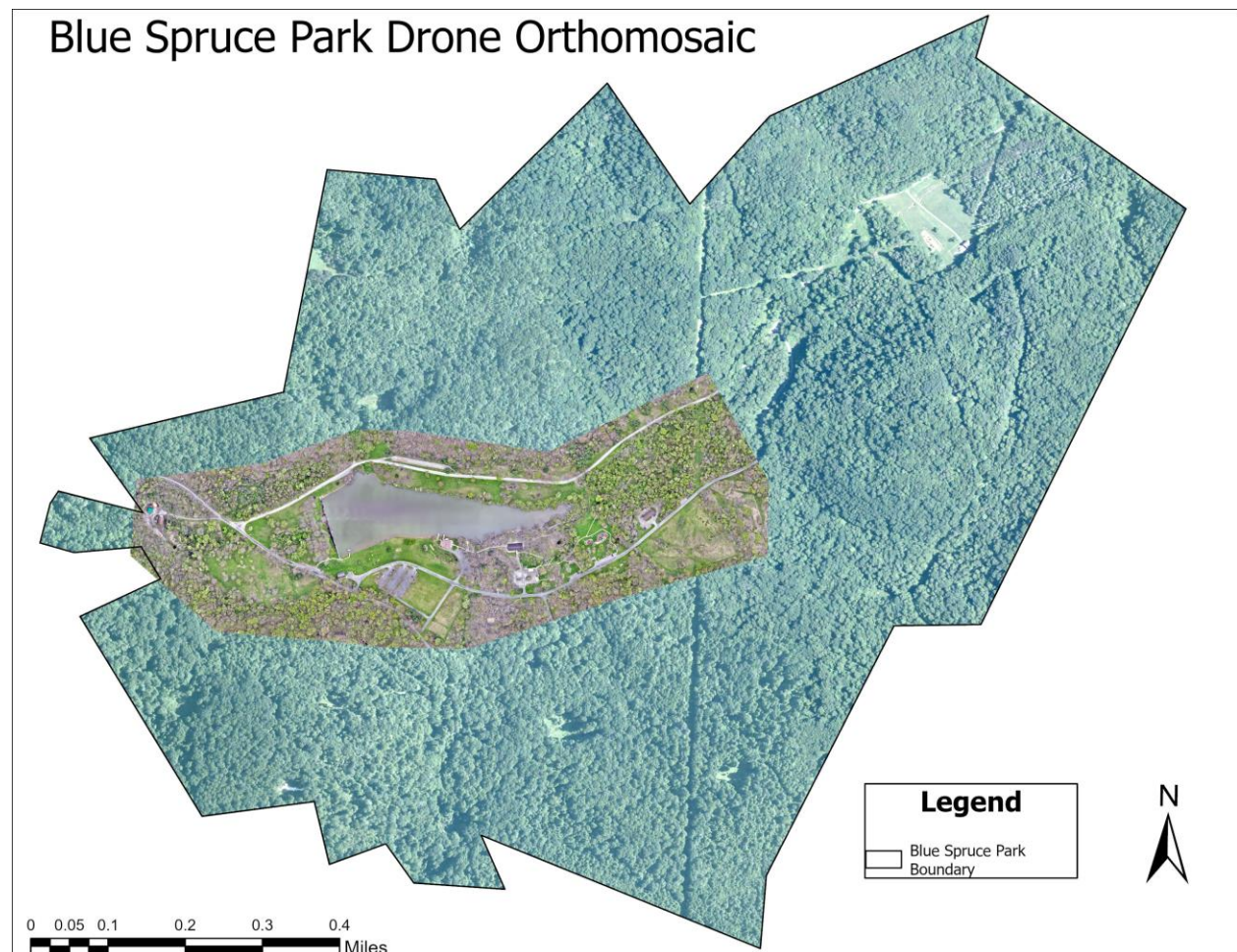
Class_name

-  Water
-  Developed
-  Forest
-  Herbaceous
-  Planted / Cultivated

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Final map with map properties

UAV data

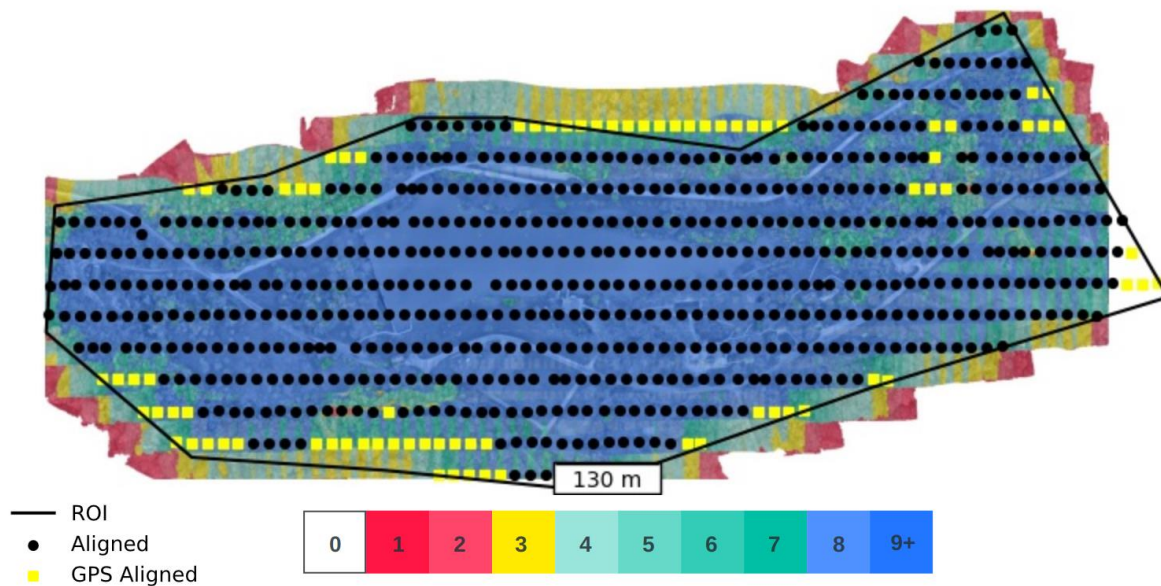


Drone Orthomosaic

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Orthomosaic Coverage



Insufficient coverage, expect large holes in the map, and low accuracy.

Marginal coverage, expect distortion or holes on buildings or sharp edges, and lower accuracy measurements.

Good coverage, expect a high quality reconstruction

Sensor(s) Used	DJI - M3E
Image Count (by sensor)	664
Image Resolution	5280x3956 (~21MP)
Orthomosaic coverage (% of area of interest)	114.60
Average Orthomosaic Image Density within Structured Area	9 images/pixel
Median Shutter Speed	1/250

Map Summery

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Project Name	Blue Spruce Cummings Reservoir - Map Plan
Photogrammetry Engine	DroneDeploy Proprietary
Processing Mode	Standard
Date Of Capture	May 01, 2026
Date Processed	May 01, 2026
GSD Orthomosaic (GSD DEM)	0.78in/px (DEM 3.12in/px)
Area Bounds (Coverage)	5514247.79ft ² (114%)
Image Sensors	DJI - M3E
Average GPS Trust	32.81ft

Quality and Accuracy

Image Quality	High texture images
Median Shutter Speed	1/250
Images Uploaded (Aligned %)	664 (88%)
Camera Optimization	0.01% variation from reference intrinsics

Discussion- Keith, Cami

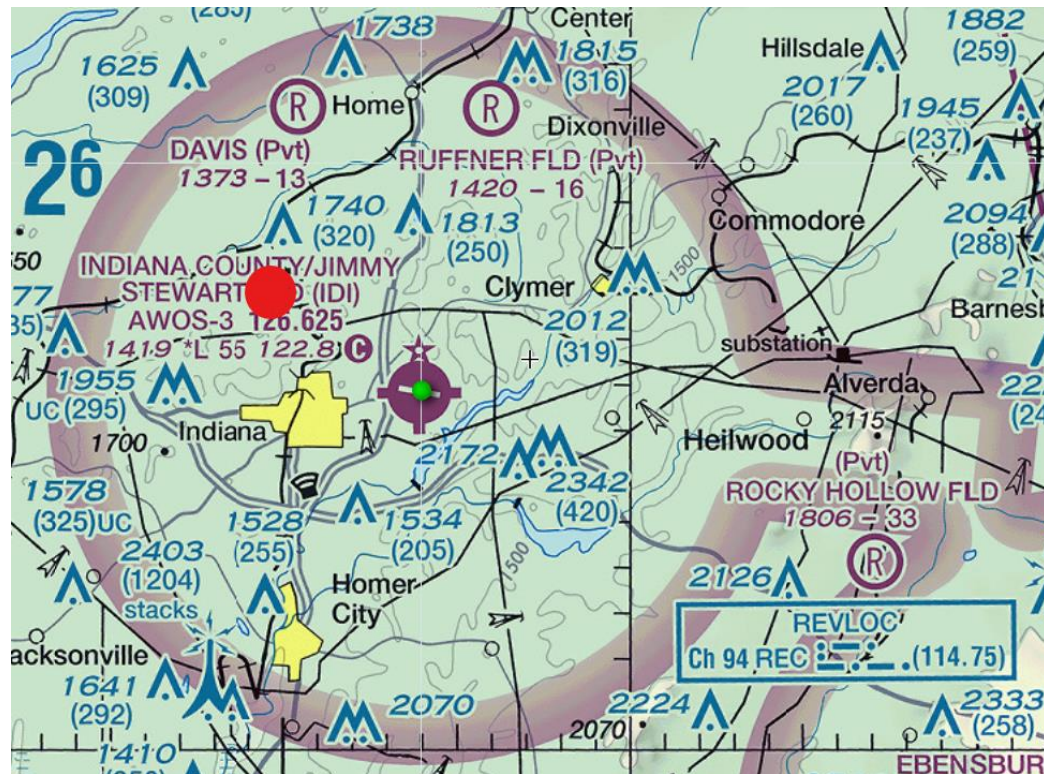
UAV issues

The original maps of Blue Spruce Park contained numerous inaccuracies. The map had been made years ago and hadn't been updated with the new trails and features that have been added to the park in the following years. The purpose of this project was to use UAV's and GPS technology to properly record the locations of all the trails in Blue Spruce Park.

Imagery can only be captured when weather conditions permit; high winds can cause drones to struggle to maintain their flight paths or even potentially crash. The turbulence can also cause distortions in captured imagery as the drone moves side to side to maintain its flight path. Heavy rain also isn't ideal due to the possibility for lenses to get wet and potential water damage to sensitive components. Due to these weather hazards it's important to only conduct flights on clear sunny days to produce the highest quality images possible. It's illegal to fly without following FAA Part 107 regulations regarding UAV flights. When flying a UAV, you must have someone maintaining eye contact with the UAV at all times. To ensure compliance with Part 107, Dr. Chris Schaney acted as our Remote Pilot in Command with students acting as visual observers spread around the park to ensure that eye contact was maintained throughout the duration of the flight. Our flight mission could easily be conducted under 400ft above ground level, so there was no need to apply for a FAA waiver to bypass Part 107 restrictions. In the event of an airplane or

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helicopter entering our flight area, we were prepared to abruptly end our flight to allow the manned aircraft to pass.



As shown in the figure above, there were no airspace restrictions during flight. The red circle marks Blue Spruce Park, which is within the Class E airspace surrounding Indiana County/Jimmy Stewart Airport. The park is 7 miles away from the airport, and was no factor in pattern operations. Prior permission to fly in the area was granted by Indiana County Parks and Trails. While the park boundary spans 650 acres, only common areas were flown and captured due to its larger area. This was a result of the topography, as there was the possibility of losing connection and line of sight with the drone due to the nearby ridge and tall trees. These factors made choosing a preferred drone for this project important. Due to the size of the park, the Mavic 3 drone could not capture the whole boundary in an efficient amount of time because of battery life, as well as it possesses an inferior camera to the Wingtra. The battery life of a Mavic 3 is typically around 30-40 minutes, and the size of the park would require four separate flights to cover the entire boundary. To compensate for the time restrictions with the Mavic 3, the boundary captured was decreased to strictly common areas. This significantly altered the required flight time.

The Wingtra Gen II drone would more efficiently cover the area of the park in one flight. It is much larger, and its bright orange color makes it easier to see through terrain. The Wingtra

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Gen II is the preferred drone for map revisions at this scale and should be used for future park revisions.

For the type of project we were doing; creating a map of a park, the GPS accuracy afforded by the surveying sticks and drones was more than enough to get accurate data to create a trail map. Inaccuracies within feet of the actual trails and boundaries aren't critical when dealing with recreation trails in a forest. Had we been conducting survey work for a building construction site than we'd pay more heed to potential inaccuracy.